

AI-ASSISTANT-CODING-LAB-14.4

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Task 1: Responsive Homepage Layout for a Startup Website

Task Description

Create a basic responsive homepage structure for a startup including a header with navigation, a main content section, and a footer. The layout must adapt smoothly across desktop, tablet, and mobile devices using CSS media queries.

Prompt

"Act as a Senior Frontend Developer. Generate a single-file HTML solution (including CSS) for a startup homepage called 'TechNova'. Include a sticky header with a logo and nav links (Home, About, Services, Contact), a hero section with a heading and description, a features grid, and a simple footer. Use CSS media queries to make it fully responsive for mobile, tablet, and desktop."

Justification

I used a **Mobile-First mindset combined with CSS Grid and Flexbox**. Grid is ideal for the feature section because it allows easy column adjustments through media queries, while Flexbox is efficient for aligning elements inside the header. The sticky header improves user experience by keeping navigation visible even when scrolling. Media queries ensure the layout adapts properly across desktop, tablet, and mobile screens.

Task 2: Interactive Call-to-Action Button

Task Description

Add an interactive button that triggers a JavaScript-driven response (like a custom message or alert) without reloading the page.

Prompt

"Generate an HTML button styled with CSS to look like a call-to-action (CTA). Write a JavaScript function that triggers when the button is clicked to show a hidden message on the screen instead of using a standard alert. Include detailed comments explaining the JS logic."

Justification

Instead of using the traditional **alert()**, I implemented **DOM manipulation** to reveal a hidden message on the page. This approach is more user-friendly because it keeps the user within the page flow without interrupting their experience. Additionally, the button becomes disabled after clicking to prevent multiple submissions and provide clear visual feedback.

Task 3: Contact Form with Client-Side Validation

Task Description

Develop a contact form with fields for **Name, Email, and Message**. Use JavaScript to validate that the fields are not empty and that the email format is correct before allowing submission.

Prompt

"Create a contact form with fields for Name, Email, and Message. Add a submit button. Use JavaScript to perform client-side validation: ensure no field is empty and the email contains an '@' and a domain. Display error messages in red text below the fields if validation fails."

Justification

Client-side validation improves **user experience and efficiency** by detecting errors before sending data to the server. I used **JavaScript with regular expressions (Regex)** to check the email format and conditions to verify that all fields are filled. Error messages are displayed directly below each input field, giving clear feedback to the user and reducing unnecessary server processing.

Task 4: Dynamic List Management for a Product Page

Task Description

Create an HTML list where users can dynamically add a new item or remove the last item using JavaScript.

Prompt

"Create a simple product list using an HTML list. Add two buttons: 'Add Item' and 'Remove Last Item'. Use JavaScript to dynamically append a new list item with the text 'Product [X]' and remove the last element when the respective buttons are clicked."

Justification

This task demonstrates **dynamic DOM manipulation**. I used **createElement()** and **appendChild()** to add items to the list, which is a safe and efficient approach compared to using innerHTML. The removal functionality checks if the list contains elements before deleting the last item, preventing runtime errors.

Task 5: Modal Popup for User Notifications

Task Description

Design a modal notification popup that appears over the screen content. It must include a semi-transparent background and multiple ways to close it.

Prompt

"Build a modal popup using HTML/CSS/JS. The modal should have a dark, semi-transparent overlay. Include an 'Open Notification' button. The modal should be closable by clicking a 'Close' button inside the modal OR by clicking anywhere on the background overlay."

Justification

The modal is designed using **position: fixed** to ensure it stays centered and visible regardless of scrolling. The semi-transparent overlay created with **RGBA** darkens the background, helping the modal stand out and focus user attention. Multiple closing options (button inside the modal and clicking outside on the overlay) follow common **modern UX patterns**, improving usability.

CODE:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Lab 14: Web Frontend Development - AI-Assisted HTML/CSS/JS</title>

  <style>

    /* ===== GLOBAL STYLES ===== */

    * {
```

```
margin: 0;

padding: 0;

box-sizing: border-box;

}
```

```
body {

    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;

    line-height: 1.6;

    color: #333;

    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);

    min-height: 100vh;

}
```

```
.container {

    max-width: 1200px;

    margin: 0 auto;

    padding: 20px;

}
```

```
/* ===== TASK 1: RESPONSIVE HOMEPAGE LAYOUT ===== */
```

```
/* Header Styles */
```

```
header {

    background: rgba(255, 255, 255, 0.95);

    box-shadow: 0 2px 10px rgba(0, 0, 0, 0.1);

    position: sticky;

    top: 0;

    z-index: 1000;
```

```
}
```

```
nav {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  padding: 1rem 2rem;  
  flex-wrap: wrap;  
}
```

```
.logo {  
  font-size: 1.5rem;  
  font-weight: bold;  
  color: #667eea;  
}
```

```
.nav-links {  
  display: flex;  
  list-style: none;  
  gap: 2rem;  
}
```

```
.nav-links a {  
  text-decoration: none;  
  color: #333;  
  font-weight: 500;  
  transition: color 0.3s ease;  
}
```

```
.nav-links a:hover {  
    color: #667eea;  
}
```

```
/* Main Content Styles */
```

```
main {  
    background: white;  
    margin: 2rem auto;  
    padding: 2rem;  
    border-radius: 10px;  
    box-shadow: 0 5px 20px rgba(0, 0, 0, 0.2);  
}
```

```
.section {  
    margin-bottom: 3rem;  
    padding: 2rem;  
    border-radius: 8px;  
    background: #f9f9f9;  
}
```

```
.section h2 {  
    color: #667eea;  
    margin-bottom: 1rem;  
    padding-bottom: 0.5rem;  
    border-bottom: 3px solid #667eea;  
}
```

```
.section h3 {  
    color: #764ba2;  
    margin: 1.5rem 0 1rem;  
}
```

```
/* Footer Styles */
```

```
footer {  
    background: rgba(255, 255, 255, 0.95);  
    text-align: center;  
    padding: 2rem;  
    margin-top: 2rem;  
}
```

```
.footer-content {  
    display: flex;  
    justify-content: space-around;  
    flex-wrap: wrap;  
    gap: 2rem;  
    margin-bottom: 1rem;  
}
```

```
.footer-section {  
    flex: 1;  
    min-width: 200px;  
}
```

```
.footer-section h4 {  
    color: #667eea;
```

```
margin-bottom: 0.5rem;  
}
```

```
/* ===== TASK 2: INTERACTIVE CALL-TO-ACTION BUTTON ===== */
```

```
.cta-button {  
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);  
  color: white;  
  padding: 15px 40px;  
  font-size: 1.1rem;  
  border: none;  
  border-radius: 50px;  
  cursor: pointer;  
  transition: all 0.3s ease;  
  box-shadow: 0 4px 15px rgba(102, 126, 234, 0.4);  
  font-weight: 600;  
}
```

```
.cta-button:hover {  
  transform: translateY(-2px);  
  box-shadow: 0 6px 20px rgba(102, 126, 234, 0.6);  
}
```

```
.cta-button:active {  
  transform: translateY(0);  
}
```

```
.message-box {
```



```
display: none;

margin-top: 1rem;

padding: 1rem;

background: #d4edda;

color: #155724;

border: 1px solid #c3e6cb;

border-radius: 5px;

animation: slideDown 0.3s ease;

}
```

```
@keyframes slideDown {

  from {

    opacity: 0;

    transform: translateY(-10px);

  }

  to {

    opacity: 1;

    transform: translateY(0);

  }

}
```

```
/* ===== TASK 3: CONTACT FORM WITH VALIDATION ===== */
```

```
.contact-form {

  max-width: 600px;

  margin: 2rem auto;

}
```

```
.form-group {  
  margin-bottom: 1.5rem;  
}
```

```
.form-group label {  
  display: block;  
  margin-bottom: 0.5rem;  
  font-weight: 600;  
  color: #333;  
}
```

```
.form-group input,  
.form-group textarea {  
  width: 100%;  
  padding: 12px;  
  border: 2px solid #ddd;  
  border-radius: 5px;  
  font-size: 1rem;  
  transition: border-color 0.3s ease;  
}
```

```
.form-group input:focus,  
.form-group textarea:focus {  
  outline: none;  
  border-color: #667eea;  
}
```

```
.form-group textarea {
```

```
    resize: vertical;
    min-height: 120px;
}
```

```
.error-message {
    color: #dc3545;
    font-size: 0.9rem;
    margin-top: 0.3rem;
    display: none;
}
```

```
.error-message.show {
    display: block;
}
```

```
.form-group input.error,
.form-group textarea.error {
    border-color: #dc3545;
}
```

```
.success-message {
    display: none;
    padding: 1rem;
    background: #d4edda;
    color: #155724;
    border: 1px solid #c3e6cb;
    border-radius: 5px;
    margin-top: 1rem;
}
```

```
}
```

```
.success-message.show {
```

```
    display: block;
```

```
}
```

```
.submit-btn {
```

```
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
```

```
    color: white;
```

```
    padding: 12px 30px;
```

```
    border: none;
```

```
    border-radius: 5px;
```

```
    cursor: pointer;
```

```
    font-size: 1rem;
```

```
    font-weight: 600;
```

```
    transition: all 0.3s ease;
```

```
}
```

```
.submit-btn:hover {
```

```
    transform: translateY(-2px);
```

```
    box-shadow: 0 4px 12px rgba(102, 126, 234, 0.4);
```

```
}
```

```
/* ===== TASK 4: DYNAMIC LIST MANAGEMENT ===== */
```

```
.product-list-container {
```

```
    max-width: 600px;
```

```
    margin: 2rem auto;
```

```
}
```

```
.product-input {  
  display: flex;  
  gap: 1rem;  
  margin-bottom: 1rem;  
}
```

```
.product-input input {  
  flex: 1;  
  padding: 10px;  
  border: 2px solid #ddd;  
  border-radius: 5px;  
  font-size: 1rem;  
}
```

```
.add-btn,  
.remove-btn {  
  padding: 10px 20px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
  font-weight: 600;  
  transition: all 0.3s ease;  
}
```

```
.add-btn {  
  background: #28a745;
```

```
    color: white;
}
```

```
.add-btn:hover {
    background: #218838;
    transform: translateY(-2px);
}
```

```
.remove-btn {
    background: #dc3545;
    color: white;
}
```

```
.remove-btn:hover {
    background: #c82333;
    transform: translateY(-2px);
}
```

```
.product-list {
    list-style: none;
    padding: 0;
}
```

```
.product-list li {
    background: white;
    padding: 15px;
    margin-bottom: 10px;
    border-radius: 5px;
}
```

```
border-left: 4px solid #667eea;  
box-shadow: 0 2px 5px rgba(0, 0, 0, 0.1);  
animation: fadeIn 0.3s ease;  
}
```

```
@keyframes fadeIn {  
  from {  
    opacity: 0;  
    transform: translateX(-20px);  
  }  
  to {  
    opacity: 1;  
    transform: translateX(0);  
  }  
}
```

```
/* ===== TASK 5: MODAL POPUP ===== */
```

```
.modal-trigger {  
  background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%);  
  color: white;  
  padding: 12px 30px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
  font-size: 1rem;  
  font-weight: 600;  
  transition: all 0.3s ease;
```

```
}
```

```
.modal-trigger:hover {  
    transform: translateY(-2px);  
    box-shadow: 0 4px 12px rgba(245, 87, 108, 0.4);  
}
```

```
.modal-overlay {  
    display: none;  
    position: fixed;  
    top: 0;  
    left: 0;  
    width: 100%;  
    height: 100%;  
    background: rgba(0, 0, 0, 0.7);  
    z-index: 2000;  
    justify-content: center;  
    align-items: center;  
    animation: fadeInOverlay 0.3s ease;  
}
```

```
.modal-overlay.show {  
    display: flex;  
}
```

```
@keyframes fadeInOverlay {  
    from {  
        opacity: 0;
```



```
}  
  
to{  
    opacity: 1;  
}  
}  
  
.modal{  
    background: white;  
    padding: 2rem;  
    border-radius: 10px;  
    max-width: 500px;  
    width: 90%;  
    position: relative;  
    animation: slideUp 0.3s ease;  
    box-shadow: 0 10px 40px rgba(0, 0, 0, 0.3);  
}  
  
@keyframes slideUp {  
    from {  
        transform: translateY(50px);  
        opacity: 0;  
    }  
    to{  
        transform: translateY(0);  
        opacity: 1;  
    }  
}
```

```
.modal h3 {  
    color: #667eea;  
    margin-bottom: 1rem;  
}
```

```
.modal p {  
    margin-bottom: 1.5rem;  
    line-height: 1.6;  
}
```

```
.close-btn {  
    position: absolute;  
    top: 10px;  
    right: 15px;  
    background: none;  
    border: none;  
    font-size: 2rem;  
    cursor: pointer;  
    color: #999;  
    transition: color 0.3s ease;  
}
```

```
.close-btn:hover {  
    color: #333;  
}
```

```
.modal-close-button {  
    background: #6c757d;
```

```
color: white;

padding: 10px 25px;

border: none;

border-radius: 5px;

cursor: pointer;

font-weight: 600;

transition: all 0.3s ease;
}
```

```
.modal-close-button:hover {

    background: #5a6268;

}
```

```
/* ===== RESPONSIVE DESIGN - MEDIA QUERIES ===== */
```

```
/* Tablet View (768px and below) */
```

```
@media (max-width: 768px) {

    nav {

        flex-direction: column;

        gap: 1rem;

    }

}
```

```
.nav-links {

    flex-direction: column;

    gap: 1rem;

    text-align: center;

}
```

```
.container {  
    padding: 15px;  
}  
  
main {  
    padding: 1.5rem;  
}  
  
.section {  
    padding: 1.5rem;  
}  
  
.footer-content {  
    flex-direction: column;  
    text-align: center;  
}  
  
.product-input {  
    flex-direction: column;  
}  
  
.modal {  
    width: 95%;  
    padding: 1.5rem;  
}  
}  
  
/* Mobile View (480px and below) */
```

```
@media (max-width: 480px) {
```

```
  .logo {
```

```
    font-size: 1.3rem;
```

```
  }
```

```
  .nav-links {
```

```
    gap: 0.5rem;
```

```
  }
```

```
  .section h2 {
```

```
    font-size: 1.5rem;
```

```
  }
```

```
  .cta-button {
```

```
    padding: 12px 30px;
```

```
    font-size: 1rem;
```

```
  }
```

```
  main {
```

```
    margin: 1rem auto;
```

```
    padding: 1rem;
```

```
  }
```

```
  .section {
```

```
    padding: 1rem;
```

```
  }
```

```
  .form-group input,
```

```
.form-group textarea {
    font-size: 0.9rem;
}
}
</style>
</head>
<body>

<!-- ===== TASK 1: RESPONSIVE HOMEPAGE LAYOUT ===== -->

<!--
AI PROMPT FOR TASK 1:
"Create a responsive homepage layout with HTML and CSS that includes:
1. A header with a logo and navigation links (Home, About, Services, Contact)
2. A main content section with clear separation
3. A footer with company information divided into sections
4. Use CSS media queries for responsive design at breakpoints: 768px (tablet)
and 480px (mobile)
5. Make the navigation collapse vertically on smaller screens
6. Ensure all elements have proper spacing and modern styling
7. Use a gradient background and white content cards with shadows"
-->

<!-- Header with Navigation -->
<header>
    <nav>
        <div class="logo">TechStartup</div>
        <ul class="nav-links">
            <li><a href="#home">Home</a></li>
```

```
<li><a href="#about">About</a></li>

<li><a href="#services">Services</a></li>

<li><a href="#contact">Contact</a></li>

</ul>

</nav>

</header>
```

```
<!-- Main Content Section -->
```

```
<main class="container">
```

```
<h1 style="text-align: center; color: #667eea; margin-bottom: 2rem;">
```

```
  Lab 14: Web Frontend Development
```

```
</h1>
```

```
<!-- Task 1: Responsive Layout Demo -->
```

```
<section class="section" id="home">
```

```
  <h2>Responsive Homepage Layout</h2>
```

```
</section>
```

```
<!-- Task 2: Interactive Call-to-Action Button -->
```

```
<!--
```

AI PROMPT FOR TASK 2:

"Create an interactive call-to-action button with the following requirements:

- 1. Design a visually appealing button with gradient background and hover effects**
- 2. Add JavaScript to handle click events without page reload**
- 3. Display a message box below the button when clicked**
- 4. Add smooth animations for the message appearance**

5. Include detailed comments in JavaScript explaining the event handling logic

6. Use modern CSS transitions and transforms for hover effects"

-->

```
<section class="section" id="cta">
```

```
  <h2>Interactive Call-to-Action Button</h2>
```

```
  <div style="text-align: center; margin-top: 2rem;">
```

```
    <button class="cta-button" id="ctaButton">Get Started Now!</button>
```

```
    <div class="message-box" id="messageBox">
```

```
      🎉 Thank you for your interest! We'll be in touch soon.
```

```
    </div>
```

```
  </div>
```

```
</section>
```

<!-- Task 3: Contact Form with Client-Side Validation -->

<!--

AI PROMPT FOR TASK 3:

"Create a contact form with client-side validation in JavaScript:

1. Include form fields for Name, Email, and Message (all required)
2. Implement validation that prevents empty submissions
3. Validate email format using regex pattern
4. Display inline error messages below each field for invalid input
5. Show visual indicators (red border) for fields with errors
6. Display a success message when form is valid and submitted
7. Reset the form after successful submission
8. Add detailed comments explaining validation logic
9. Style error messages in red and success messages in green"

-->

```
<section class="section" id="contact">
```

```
<h2>Contact Form with Client-Side Validation</h2>
```

```
<form class="contact-form" id="contactForm">
```

```
<div class="form-group">
```

```
<label for="name">Name *</label>
```

```
<input type="text" id="name" name="name" placeholder="Enter your  
name">
```

```
<span class="error-message" id="nameError">Please enter your  
name</span>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="email">Email *</label>
```

```
<input type="email" id="email" name="email" placeholder="Enter your  
email">
```

```
<span class="error-message" id="emailError">Please enter a valid email  
address</span>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="message">Message *</label>
```

```
<textarea id="message" name="message" placeholder="Enter your  
message"></textarea>
```

```
<span class="error-message" id="messageError">Please enter a  
message</span>
```

```
</div>
```

<button type="submit" class="submit-btn">Submit</button>

<div class="success-message" id="successMessage">

✓ Form submitted successfully! Thank you for contacting us.

</div>

</form>

</section>

<!-- Task 4: Dynamic List Management -->

<!--

AI PROMPT FOR TASK 4:

"Create a dynamic product list management system with JavaScript:

- 1. Display an input field for entering product names**
- 2. Add an 'Add Product' button that appends new items to the list**
- 3. Add a 'Remove Last Product' button that removes the last item**
- 4. Start with 3 pre-populated sample products**
- 5. Update the DOM dynamically without page reload**
- 6. Support Enter key functionality to add products**
- 7. Show alerts if trying to add empty products or remove from empty list**
- 8. Add smooth animations when items are added**
- 9. Include clear comments explaining DOM manipulation**
- 10. Style buttons with different colors (green for add, red for remove)"**

-->

<section class="section" id="products">

<h2>Dynamic Product List Management</h2>

```
<div class="product-list-container">

  <div class="product-input">

    <input type="text" id="productInput" placeholder="Enter product name">

    <button class="add-btn" id="addProductBtn">Add Product</button>

  </div>

  <button class="remove-btn" id="removeProductBtn">Remove Last
Product</button>
```

```

    <h3 style="margin-top: 2rem; color: #667eea;">Product List:</h3>

    <ul class="product-list" id="productList">

      <li>Laptop - Premium Model</li>

      <li>Wireless Mouse</li>

      <li>Mechanical Keyboard</li>

    </ul>

  </div>

</section>
```

<!-- Task 5: Modal Popup -->

<!--

AI PROMPT FOR TASK 5:

"Create a modal popup system with the following features:

- 1. HTML structure for a modal with overlay background**
- 2. CSS styling with semi-transparent dark overlay (70% opacity)**
- 3. Center the modal on the screen with slide-up animation**
- 4. Include an X button in the top-right corner**
- 5. Add a Close button at the bottom**
- 6. Allow closing by clicking outside the modal (on overlay)**

7. Implement JavaScript for all three closing methods
8. Add smooth fade-in animations for overlay and slide-up for modal
9. Use z-index to ensure modal appears above all content
10. Include detailed comments explaining event handling logic
11. Make modal responsive and mobile-friendly"

-->

```
<section class="section" id="modal-demo">
  <h2>Modal Popup Notifications</h2>
  <div style="text-align: center; margin-top: 2rem;">
    <button class="modal-trigger" id="openModalBtn">Open Modal</button>
  </div>
</section>
</main>

<!-- Footer -->
<footer>
  <div class="footer-content">
    <div class="footer-section">
      <h4>About Us</h4>
      <p>TechStartup - Building tomorrow's solutions today.</p>
    </div>
    <div class="footer-section">
      <h4>Contact</h4>
      <p>Email: info@techstartup.com</p>
      <p>Phone: (555) 123-4567</p>
    </div>
    <div class="footer-section">
```

<h4>Follow Us</h4>

<p>Facebook | Twitter | LinkedIn</p>

</div>

</div>

<p style="margin-top: 1rem; color: #666;">

© 2026 TechStartup. All rights reserved.

</p>

</footer>

<!-- Modal Popup Structure -->

<div class="modal-overlay" id="modalOverlay">

<div class="modal">

<button class="close-btn" id="closeModalX">×</button>

<h3> 🛎 Important Notification</h3>

<p>

**Welcome to our website! We're glad you're here. This modal demonstrates
how to create overlay popups for announcements, alerts, or important
information.**

</p>

<p>

Modal popups are great for:

</p>

<ul style="margin-left: 2rem; margin-bottom: 1.5rem;">

Displaying important announcements

Collecting user information

Showing confirmations or alerts

Focusing user attention on specific content

```
        <button class="modal-close-button" id="closeModalBtn">Close</button>

    </div>

</div>
```

```
<script>
```

```
    // ===== TASK 2: INTERACTIVE CALL-TO-ACTION BUTTON =====

    // AI PROMPT: "Write JavaScript to handle button clicks and display a message
    without alert popups"
```

```
    /**
```

```
    * Event listener for the CTA button
```

```
    * When clicked, displays a message box below the button
```

```
    * Demonstrates JavaScript event handling and DOM manipulation
```

```
    */
```

```
    document.getElementById('ctaButton').addEventListener('click', function() {
```

```
        // Get reference to the message box element
```

```
        const messageBox = document.getElementById('messageBox');
```

```
        // Toggle display of the message box
```

```
        if (messageBox.style.display === 'none' || messageBox.style.display === '') {
```

```
            messageBox.style.display = 'block';
```

```
        } else {
```

```
            messageBox.style.display = 'none';
```

```
        }
```

```
        // Optional: Display an alert as well
```

```
        // Uncomment the line below if you want an alert
```

```
        // alert('Thank you for clicking! We appreciate your interest.');
```

```
});
```

```
// ===== TASK 3: CONTACT FORM WITH CLIENT-SIDE VALIDATION =====
```

```
// AI PROMPT: "Write form validation JavaScript with regex for email and inline  
error display"
```

```
/**
```

```
 * Form validation function
```

```
 * Validates form inputs before submission
```

```
 * Prevents empty submissions and checks email format
```

```
 */
```

```
document.getElementById('contactForm').addEventListener('submit',  
function(event) {
```

```
    // Prevent default form submission
```

```
    event.preventDefault();
```

```
    // Get form field values
```

```
    const name = document.getElementById('name').value.trim();
```

```
    const email = document.getElementById('email').value.trim();
```

```
    const message = document.getElementById('message').value.trim();
```

```
    // Get error message elements
```

```
    const nameError = document.getElementById('nameError');
```

```
    const emailError = document.getElementById('emailError');
```

```
    const messageError = document.getElementById('messageError');
```

```
    // Get input elements
```

```
    const nameInput = document.getElementById('name');
```

```
    const emailInput = document.getElementById('email');
```

```
const messageInput = document.getElementById('message');
```

```
// Reset previous error states
```

```
nameError.classList.remove('show');
```

```
emailError.classList.remove('show');
```

```
messageError.classList.remove('show');
```

```
nameInput.classList.remove('error');
```

```
emailInput.classList.remove('error');
```

```
messageInput.classList.remove('error');
```

```
// Validation flag
```

```
let isValid = true;
```

```
// Validate name field (cannot be empty)
```

```
if (name === '') {
```

```
    nameError.classList.add('show');
```

```
    nameInput.classList.add('error');
```

```
    isValid = false;
```

```
}
```

```
// Validate email field (cannot be empty and must be valid format)
```

```
// Regular expression for email validation
```

```
const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
if (email === '') {
```

```
    emailError.textContent = 'Please enter your email address';
```

```
    emailError.classList.add('show');
```

```
    emailInput.classList.add('error');
```

```
    isValid = false;
```



```
} else if (!emailRegex.test(email)) {  
    emailError.textContent = 'Please enter a valid email address';  
    emailError.classList.add('show');  
    emailInput.classList.add('error');  
    isValid = false;  
}  
  
// Validate message field (cannot be empty)  
if (message === '') {  
    messageError.classList.add('show');  
    messageInput.classList.add('error');  
    isValid = false;  
}  
  
// If all validations pass, show success message  
if (isValid) {  
    const successMessage = document.getElementById('successMessage');  
    successMessage.classList.add('show');  
  
    // Reset form fields  
    document.getElementById('contactForm').reset();  
  
    // Hide success message after 5 seconds  
    setTimeout(function() {  
        successMessage.classList.remove('show');  
    }, 5000);  
}  
});
```

// ===== TASK 4: DYNAMIC LIST MANAGEMENT =====

// AI PROMPT: "Write JavaScript to dynamically add/remove list items with DOM manipulation"

/**

*** Add product to the list**

*** Takes input from the text field and adds it as a new list item**

*** Updates the DOM dynamically without page reload**

***/**

**document.getElementById('addProductBtn').addEventListener('click',
function() {**

// Get the input value

const productInput = document.getElementById('productInput');

const productName = productInput.value.trim();

// Check if input is not empty

if (productName !== '') {

// Create a new list item element

const newProduct = document.createElement('li');

newProduct.textContent = productName;

// Add the new item to the product list

document.getElementById('productList').appendChild(newProduct);

// Clear the input field

productInput.value = '';

// Focus back on input for easy multiple additions

```
        productInput.focus();
    } else {
        // Alert user if trying to add empty product
        alert('Please enter a product name before adding.');
```



```
    }
});

/**
 * Remove the last product from the list
 * Removes the last child element from the product list
 * Prevents removal if list is empty
 */
document.getElementById('removeProductBtn').addEventListener('click',
function() {
    // Get reference to the product list
    const productList = document.getElementById('productList');

    // Check if there are any products in the list
    if (productList.children.length > 0) {
        // Remove the last product
        productList.removeChild(productList.lastElementChild);
    } else {
        // Alert user if list is already empty
        alert('No products to remove!');
    }
});

/**
```

```

    * Allow adding products by pressing Enter key
    * Improves user experience by supporting keyboard interaction
    */

    document.getElementById('productInput').addEventListener('keypress',
function(event) {

    // Check if Enter key was pressed (key code 13)

    if (event.key === 'Enter') {

        // Trigger the add button click

        document.getElementById('addProductBtn').click();

    }

});

// ===== TASK 5: MODAL POPUP FUNCTIONALITY =====

// AI PROMPT: "Write JavaScript to open/close modal with multiple closing
methods (button, X, overlay)"

/**
 * Open modal popup
 * Displays the modal overlay and content
 */

document.getElementById('openModalBtn').addEventListener('click', function()
{

    // Show the modal overlay

    document.getElementById('modalOverlay').classList.add('show');

});

/**
 * Close modal using the X button
 * Hides the modal when close button is clicked

```

```

*/

document.getElementById('closeModalX').addEventListener('click', function() {

    // Hide the modal overlay

    document.getElementById('modalOverlay').classList.remove('show');

});

/**

* Close modal using the Close button

* Alternative way to close the modal

*/

document.getElementById('closeModalBtn').addEventListener('click',
function() {

    // Hide the modal overlay

    document.getElementById('modalOverlay').classList.remove('show');

});

/**

* Close modal by clicking on the overlay (outside the modal content)

* Provides intuitive way to dismiss the modal

*/

document.getElementById('modalOverlay').addEventListener('click',
function(event) {

    // Check if the click was on the overlay itself, not the modal content

    if (event.target === this) {

        // Hide the modal overlay

        this.classList.remove('show');

    }

});

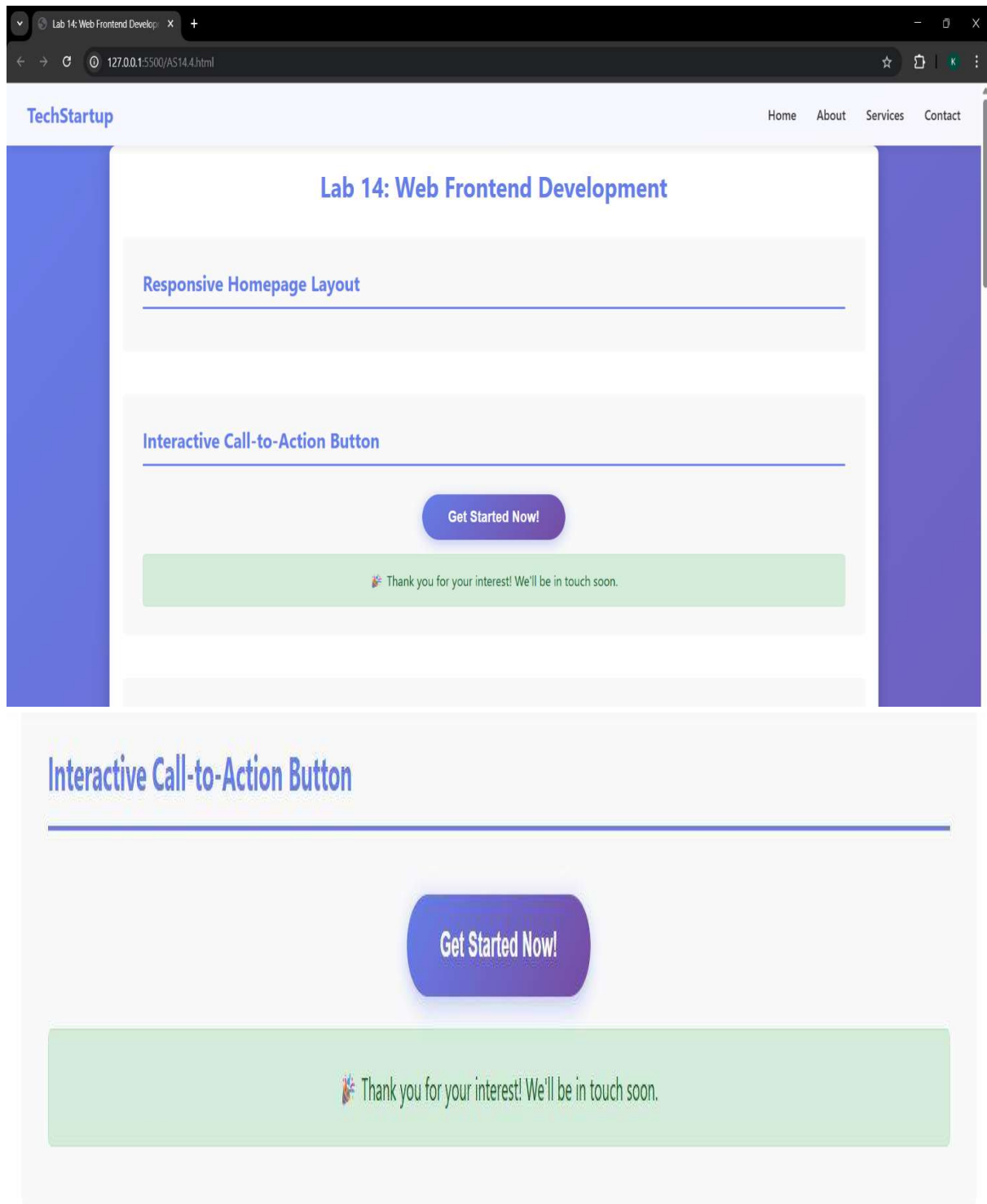
</script>

```

</body>

</html>

OUTPUT:



Contact Form with Client-Side Validation

Name *

Email *

Message *

Submit

Dynamic Product List Management

Add Product

Remove Last Product

Product List:

Laptop - Premium Model

Wireless Mouse

Mechanical Keyboard

Modal Popup Notifications

Open Modal

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